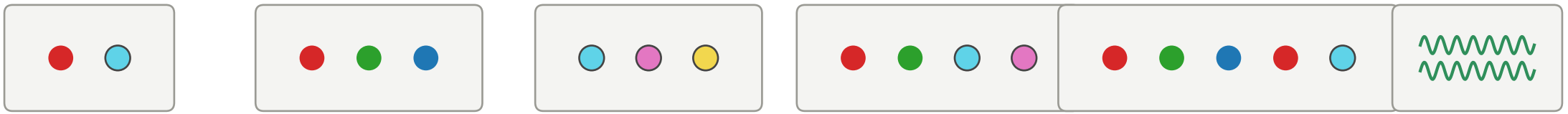
Scene 1: the guest list: which combinations of quarks QCD lets out

allowed: colour singlets, the colours cancel



meson $q\bar{q}$

$3 \otimes \bar{3} = 1 \oplus 8$

$\pi^+ (u\bar{d})$, $K^- (s\bar{u})$, $\bar{B}^{*0} (b\bar{d})$

baryon qqq

$3 \otimes 3 \otimes 3 = 1 \oplus 8 \oplus 8 \oplus 10$

$p (uud)$, $n (udd)$, $\Sigma^0 (uds)$

antibaryon $\bar{q}\bar{q}\bar{q}$

the conjugate

$\bar{n} (\bar{u}\bar{d}\bar{d})$, $\bar{\Sigma}^0 (\bar{u}\bar{d}\bar{s})$

tetraquark $qq\bar{q}\bar{q}$

rare (LHCb); not from a string

$T_{cc}^+ (cc\bar{u}\bar{d})$

pentaquark $qqqq\bar{q}$

rare (LHCb); not from a string

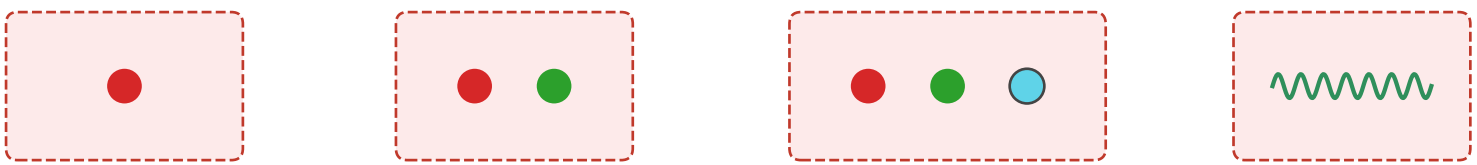
$P_c^+ (uudc\bar{c})$

glueball gg, ggg

$8 \otimes 8 \ni 1$

predicted, not yet certain

forbidden: a net colour charge, the field would stretch to infinity



a lone quark q

a triplet 3

never seen

a diquark qq

$3 \otimes 3 = \bar{3} \oplus 6$: no singlet

but a fine string end (a $\bar{3}$)

two quarks, one antiquark $qq\bar{q}$

$n_q - n_{\bar{q}} = 1$

no singlet

a lone gluon g

an octet 8

never seen

the rule

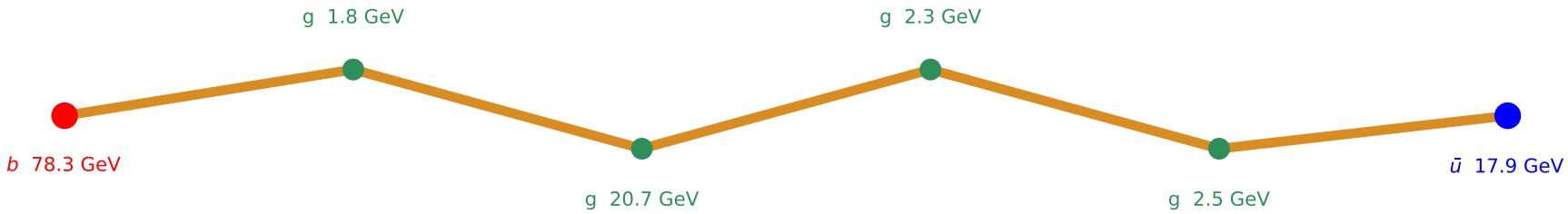
a free hadron is invariant under SU(3) of colour:
$n_q - n_{\bar{q}}$ a multiple of three, any number of gluons.
Every string break creates a pair, $q'\bar{q}'$ or a diquark pair,
so flavour, charge and baryon number are conserved,
and the string's hadrons are mesons and baryons only.

A free particle must be a colour singlet: the colours of its quarks (red, green, blue) and the anticolours of its antiquarks must cancel, otherwise the field it drags behind it would stretch to infinity. The rule is that the number of quarks minus the number of antiquarks is a multiple of three. The Lund string makes only the first two kinds, mesons and baryons: a string end is a quark, an antiquark or a diquark.

Scene 2: the wedding: the mother's rope snaps into a chain of mesons and baryons

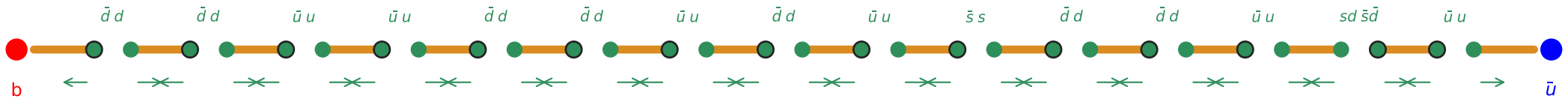
standalone_hadronisation.py

(a) the ropes of the record: the mother's, with four knots



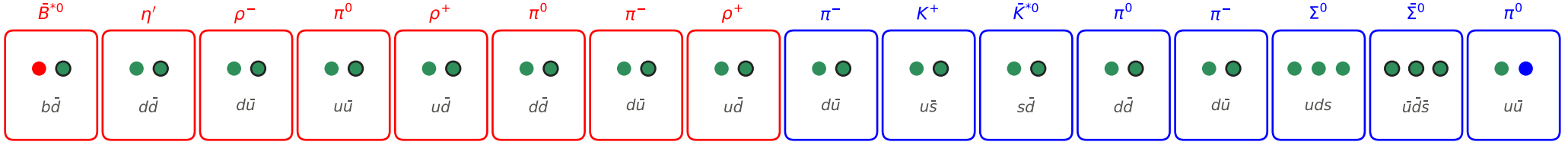
string 1 of 33 colour singlets;
the mother's b carries 78.3 GeV of p_T

(b) the rope snaps: a pair from the vacuum at every break, in the order the script drew them



15 of 15 breaks: each costs the rope's own energy, 1 GeV per fermi, and pays for one pair

(c) the procession: the 16 hadrons in their order along the rope, with their quark content

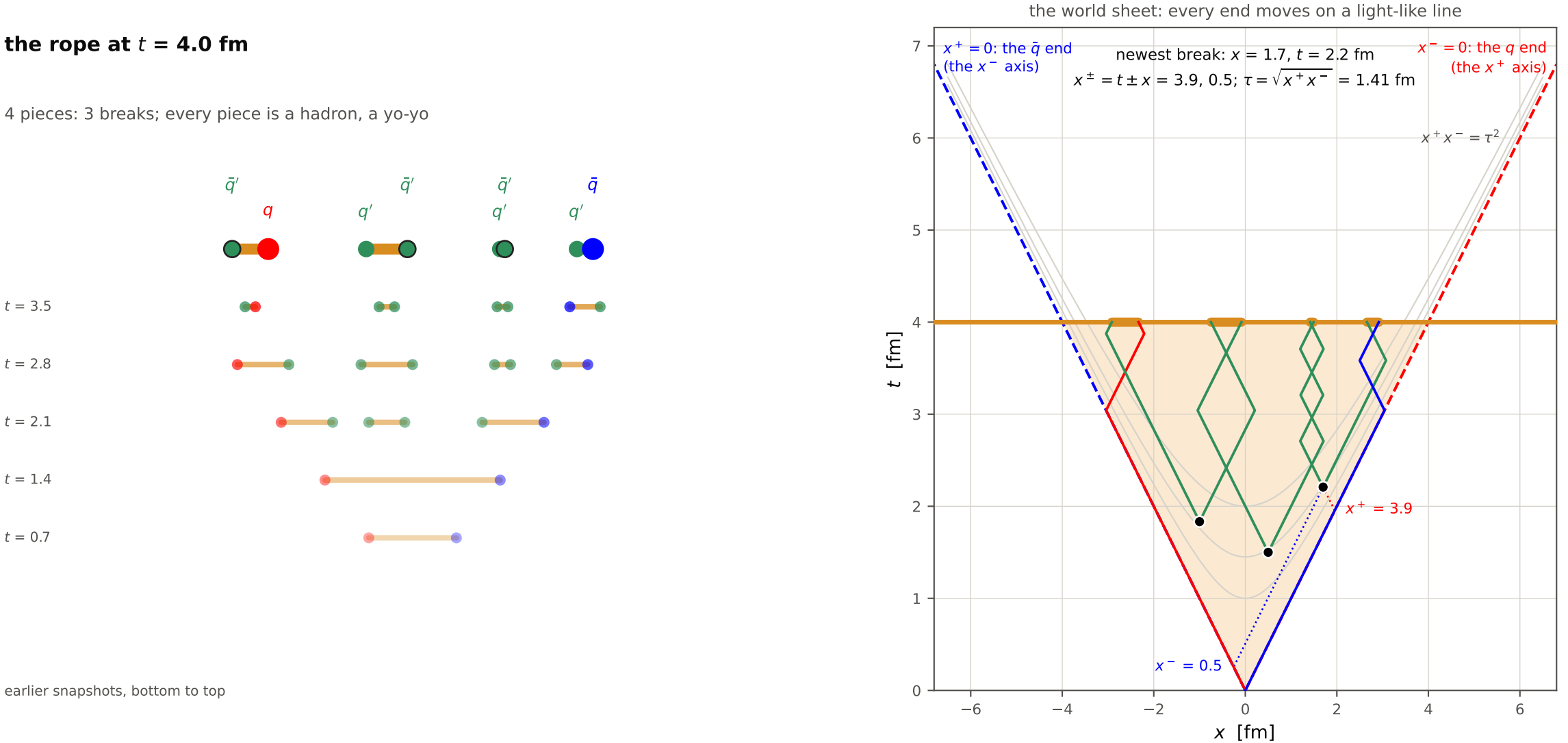


14 mesons and 2 baryons (a diquark pair makes a baryon and an antibaryon side by side); red boxes from the b end, blue from the $\bar{u}$ end; the B*bar0 ($b\bar{d}$) keeps 100% of the mother's energy

Every parton of the record holds one end of a colour line and nobody may leave alone. The mother's rope, string 1, runs from the b through 4 gluons to a $\bar{u}$ from the rest of the event (W = 69 GeV). As the partons fly apart the rope snaps, in the order the script drew, and at every break a pair from the vacuum holds the two new ends; what walks out is a chain of 16 hadrons, couples and triples, with the b in the first and the $\bar{u}$ in the last.

Scene 3: the rope in time, and why light-cone coordinates: a yo-yo at every break

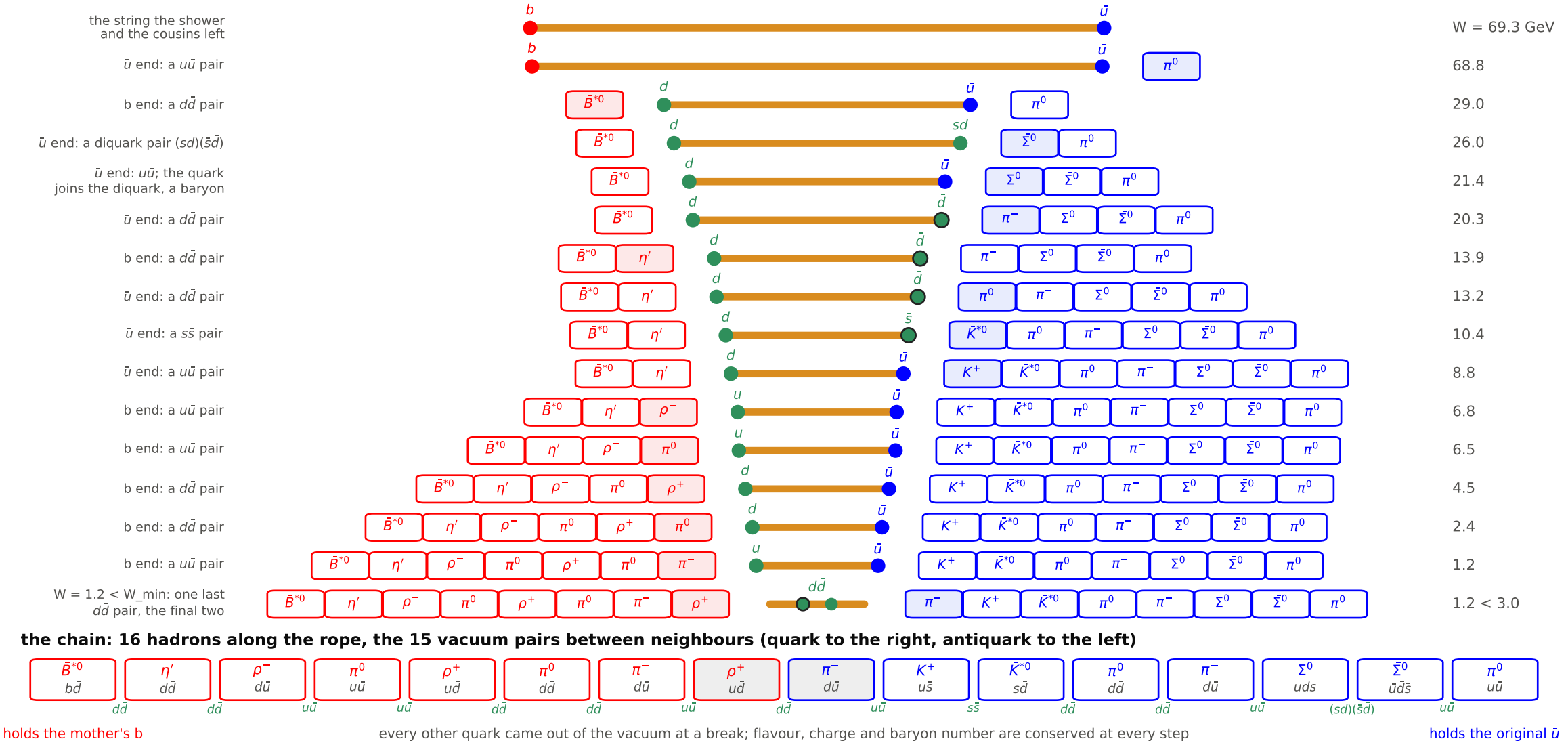
standalone_hadronisation.py



Left: a quark and an antiquark created at one point fly apart at the speed of light, the rope between them stores 1 GeV per fermi, and pairs appear at rest in the field at the break-ups; the pieces between neighbouring breaks are short strings whose ends run towards each other, cross and run apart again for ever, the yo-yos that are the hadrons. Right: the same history on the world sheet. The light-like directions are the natural axes, $x^\pm = t \pm x$: a massless end keeps one of them fixed and a break at proper time τ has $x^+ x^- = \tau^2$.

Scene 4: break by break: the mother's rope from both ends

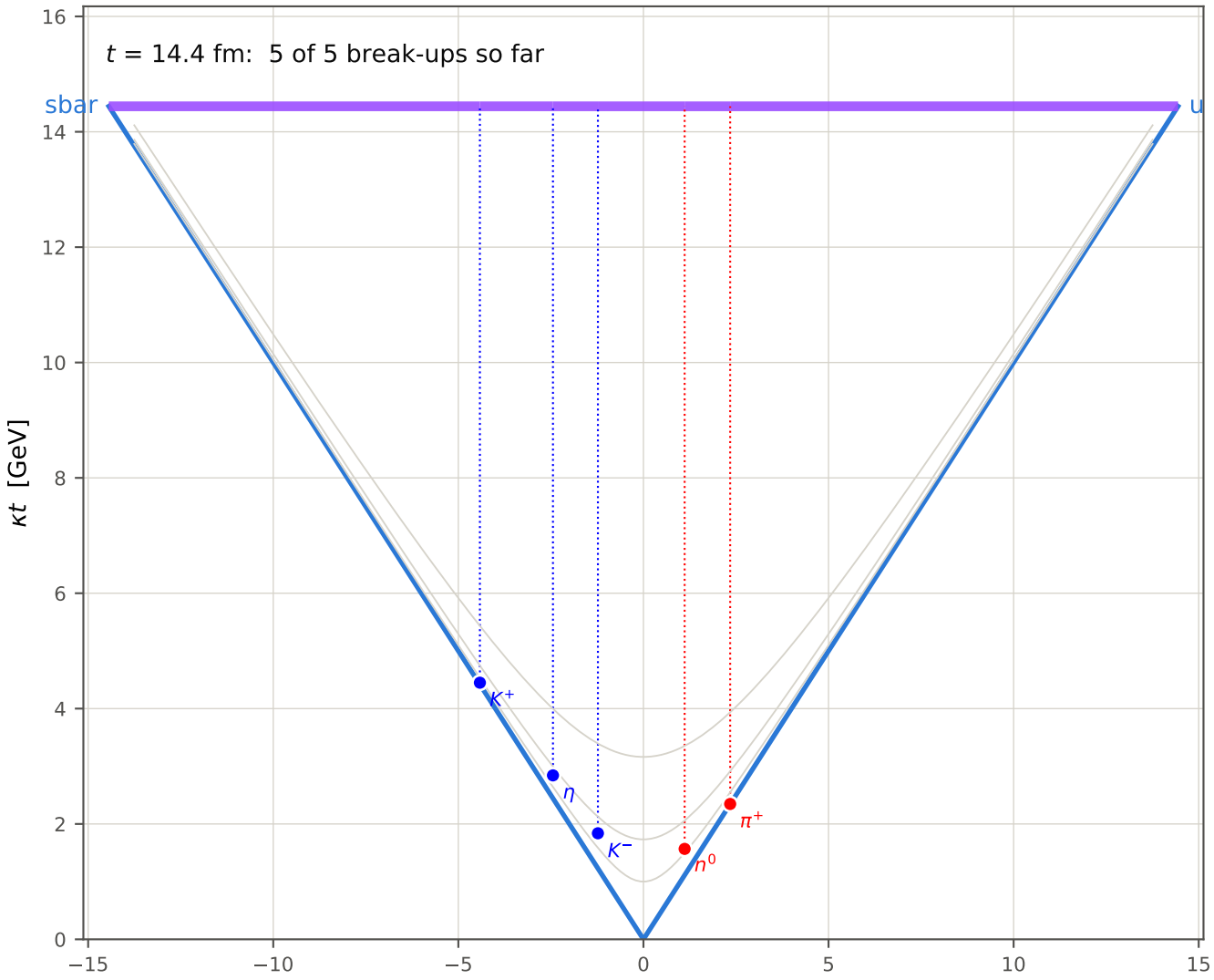
standalone_hadronisation.py

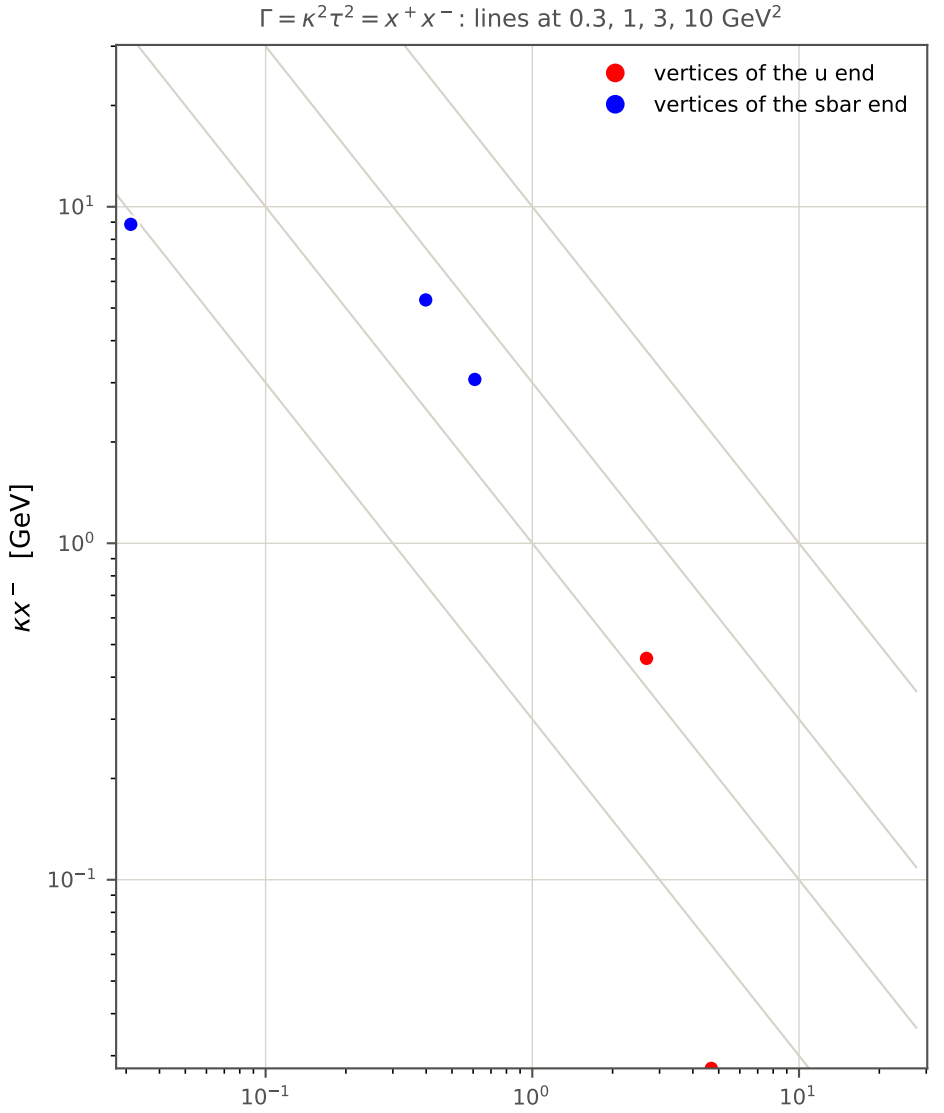


String 1 break by break, as the -v log recorded it (the gluon kinks are left out of the drawing). A break at the $\bar{u}$ end creates a $u\bar{u}$ pair: the new u joins the old $\bar{u}$ into a π^0 and the new $\bar{u}$ is the end of what is left; a break at the b end creates a $d\bar{d}$ pair and the $\bar{d}$ joins the mother. The two ends take turns in random order until the remaining mass is below W_min and the final two share what is left: 16 hadrons, 15 vacuum pairs, the b in the first, the $\bar{u}$ in the last.

Scene 5: a string breaks: the two-parton string in its rest frame

standalone_hadronisation.py

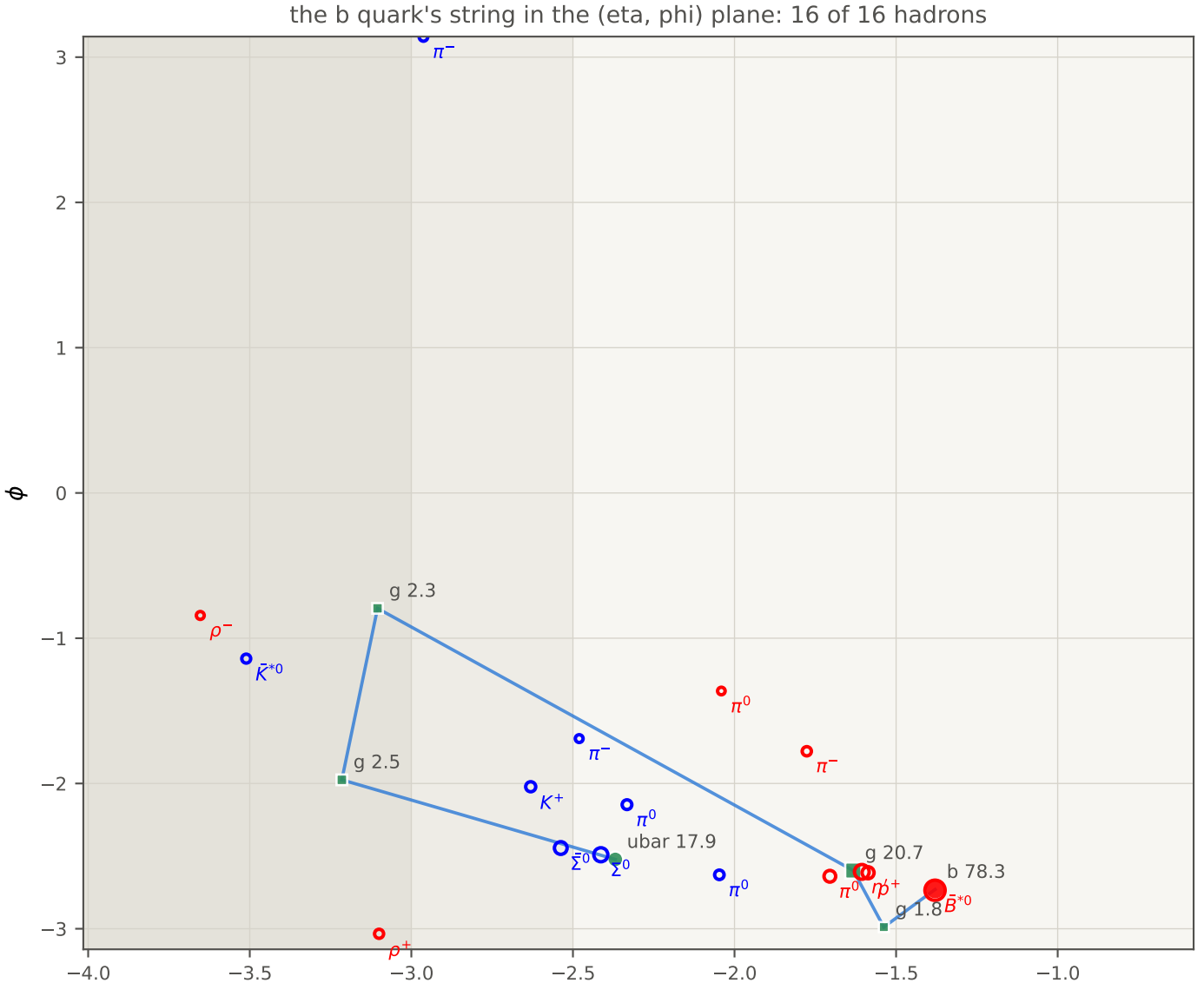




A 28 GeV u sbar string of our event in its own rest frame (seed 1). The quark and the antiquark fly apart at the speed of light, the string between them stores 1 GeV per fermi, and it breaks where the record's vertices are: every piece between two neighbouring vertices is one of the 7 hadrons. On the right the same vertices in light-cone coordinates: the area law puts them near $\Gamma = x^+x^- \sim$ 1-3 GeV^2.

Scene 6: the b quark's string: hadrons from both ends

standalone_hadronisation.py



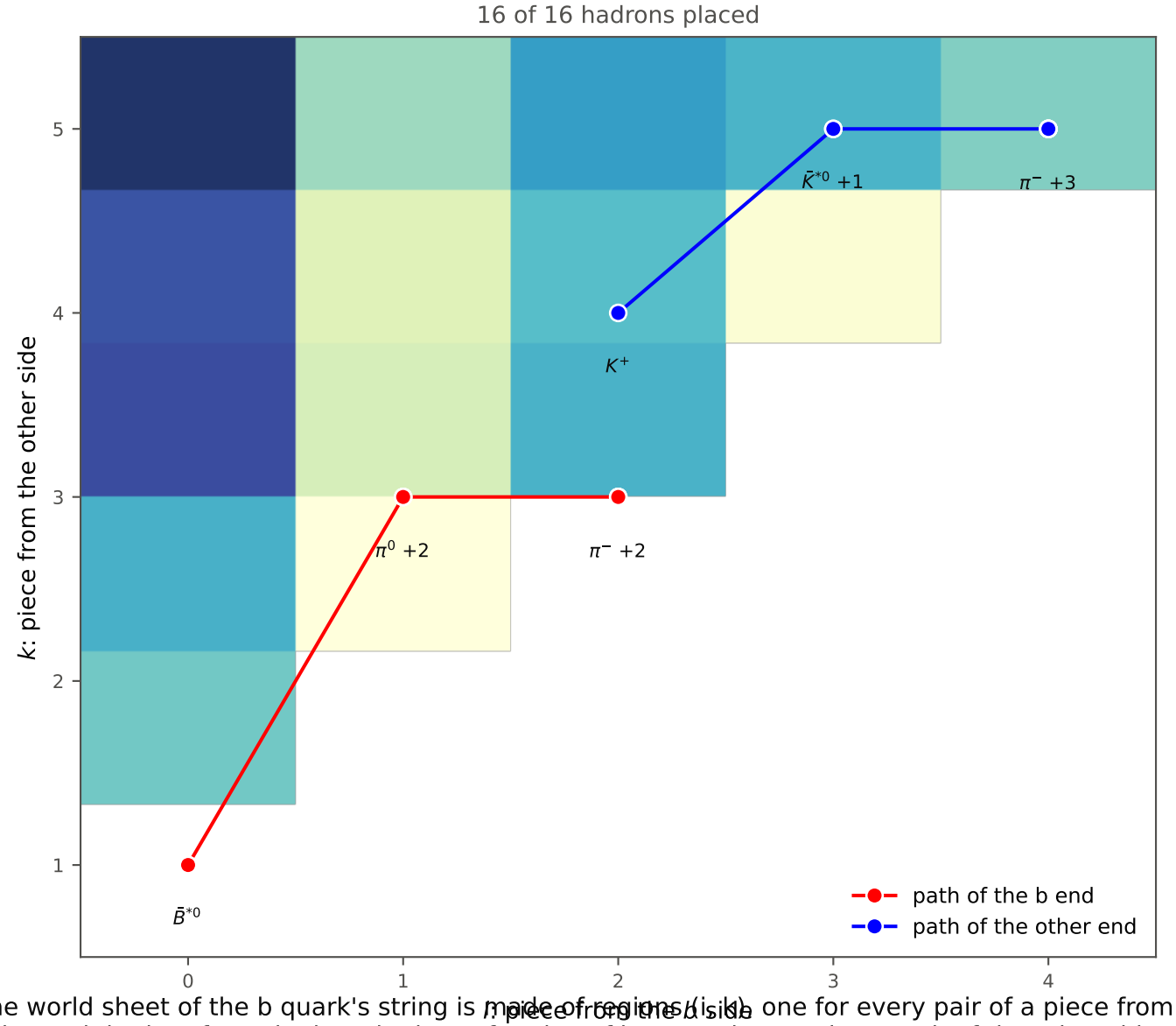
hadron	end	z	remaining W -> W_min
pi0	-	0.134	69.3 > 7.0
B*bar0	+	0.984	68.8 > 6.0
Sigmabar0	-	0.271	29.0 > 2.4
Sigma0	-	0.764	26.0 > 3.5
pi-	-	0.333	21.4 > 2.5
eta'	+	0.416	20.3 > 2.6
pi0	-	0.591	13.9 > 2.3
K*bar0	-	0.446	13.2 > 2.1
K+	-	0.565	10.4 > 2.8
rho-	+	0.314	8.8 > 2.2
pi0	+	0.242	6.8 > 2.3
rho+	+	0.444	6.5 > 2.8
pi0	+	0.603	4.5 > 2.1
pi-	+	0.274	2.4 > 2.1
rho+	+	0.549	1.2 > 3.0
pi-	-	final	1.2 > 3.0

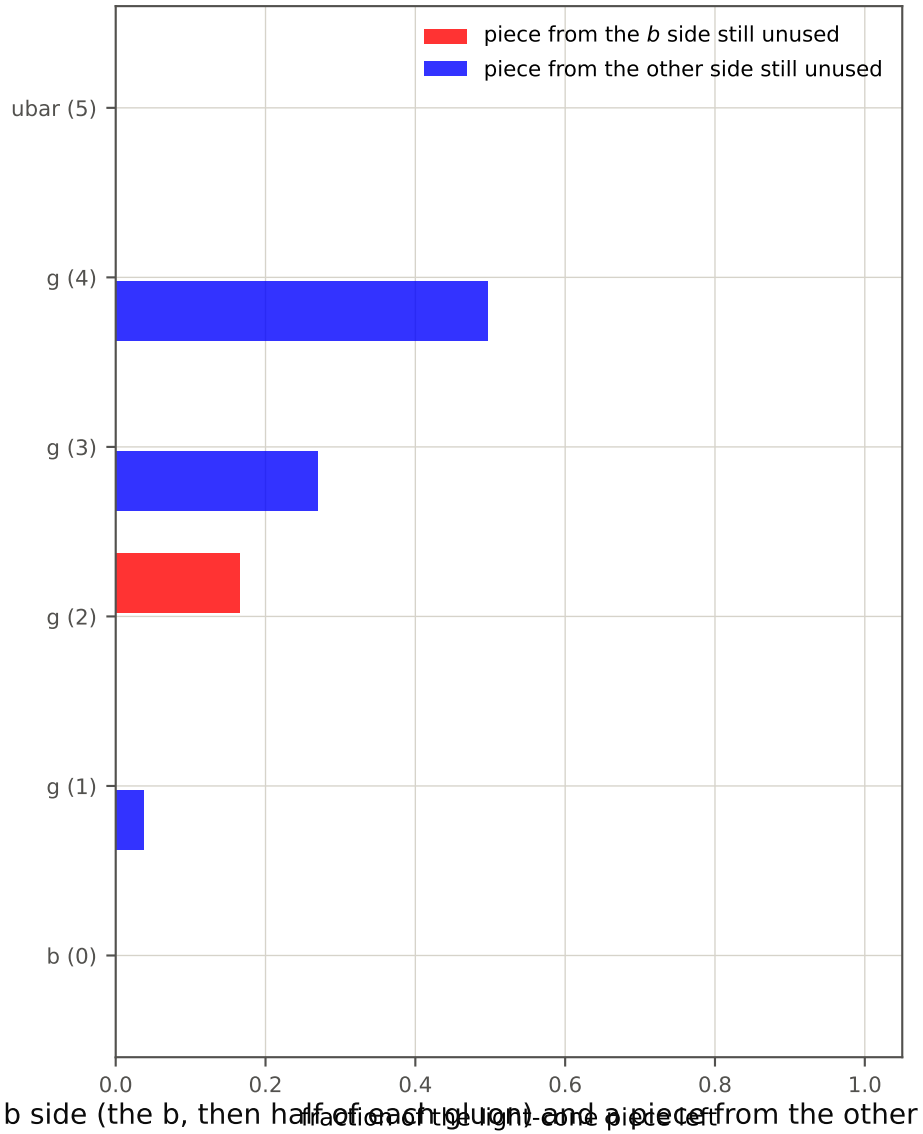
16 hadrons; momentum conserved to 1e-12 GeV;
the B*bar0 carries 100% of the b quark's energy

The string that holds the b quark: 6 partons (the b, 4 gluons and the other end), W = 69 GeV. Hadrons are produced from the two ends in random order; each takes a fraction z of the remaining light-cone momentum of its end (Lund-Bowler for the b: z = 0.984 gives the B*bar0 with x_E = 1.004) until the remaining mass drops below W_min and the last two hadrons share what is left.

Scene 7: gluons as kinks: the two ends walk through the regions of the world sheet

standalone_hadronisation.py

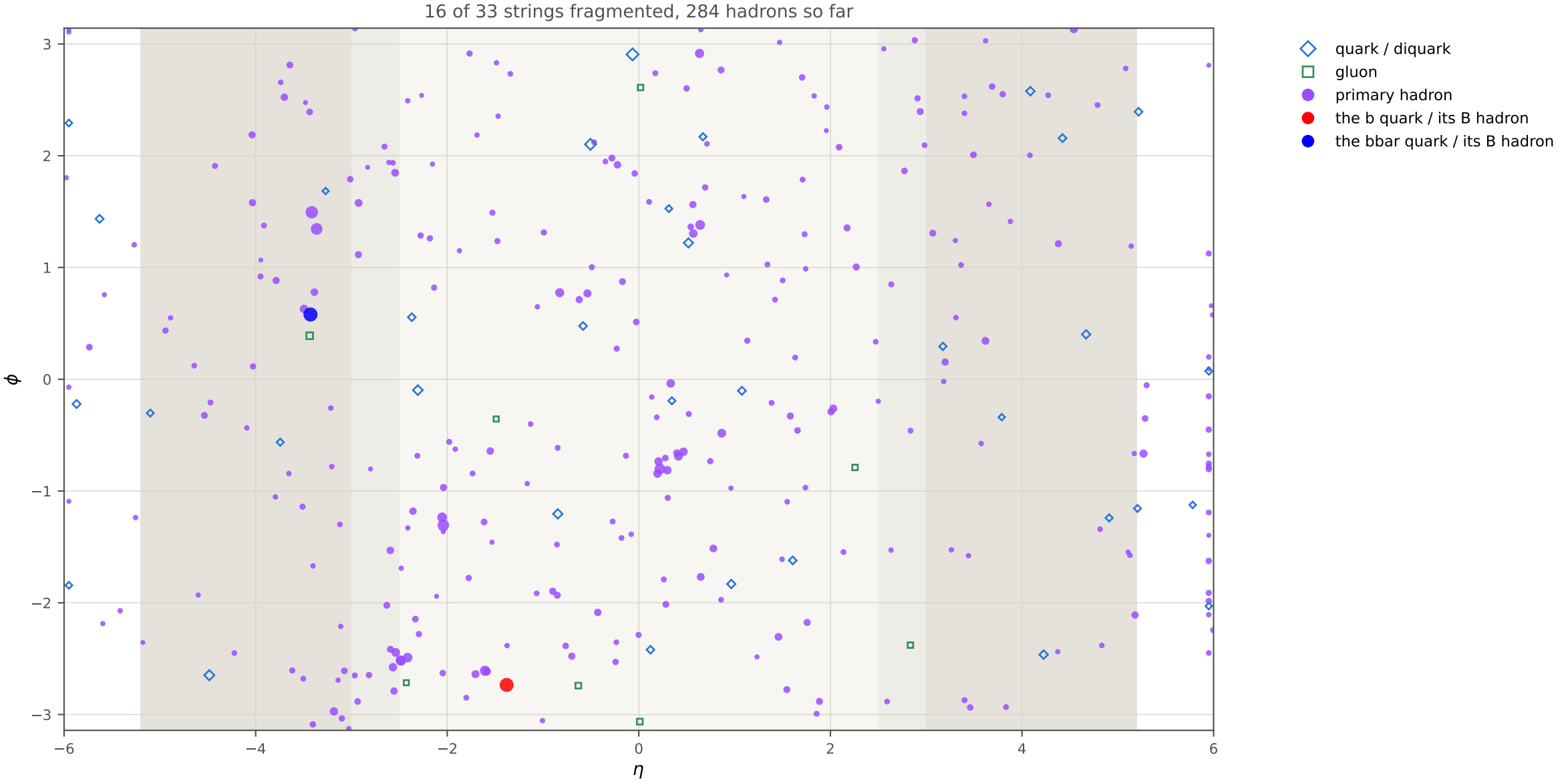




The world sheet of the b quark's string is made of regions (i, k), one for every pair of a piece from the b side (the b, then half of each gluon) and a piece from the other side. Each hadron from the b end takes a fraction of its own piece and as much of the other side as its mass needs; when a region is used up the end steps to the next one, swallowing the rest of the gluon half, which is how a hadron forms across a kink. The map darkens with the region's mass.

Scene 8: the whole event: 33 strings become 365 hadrons

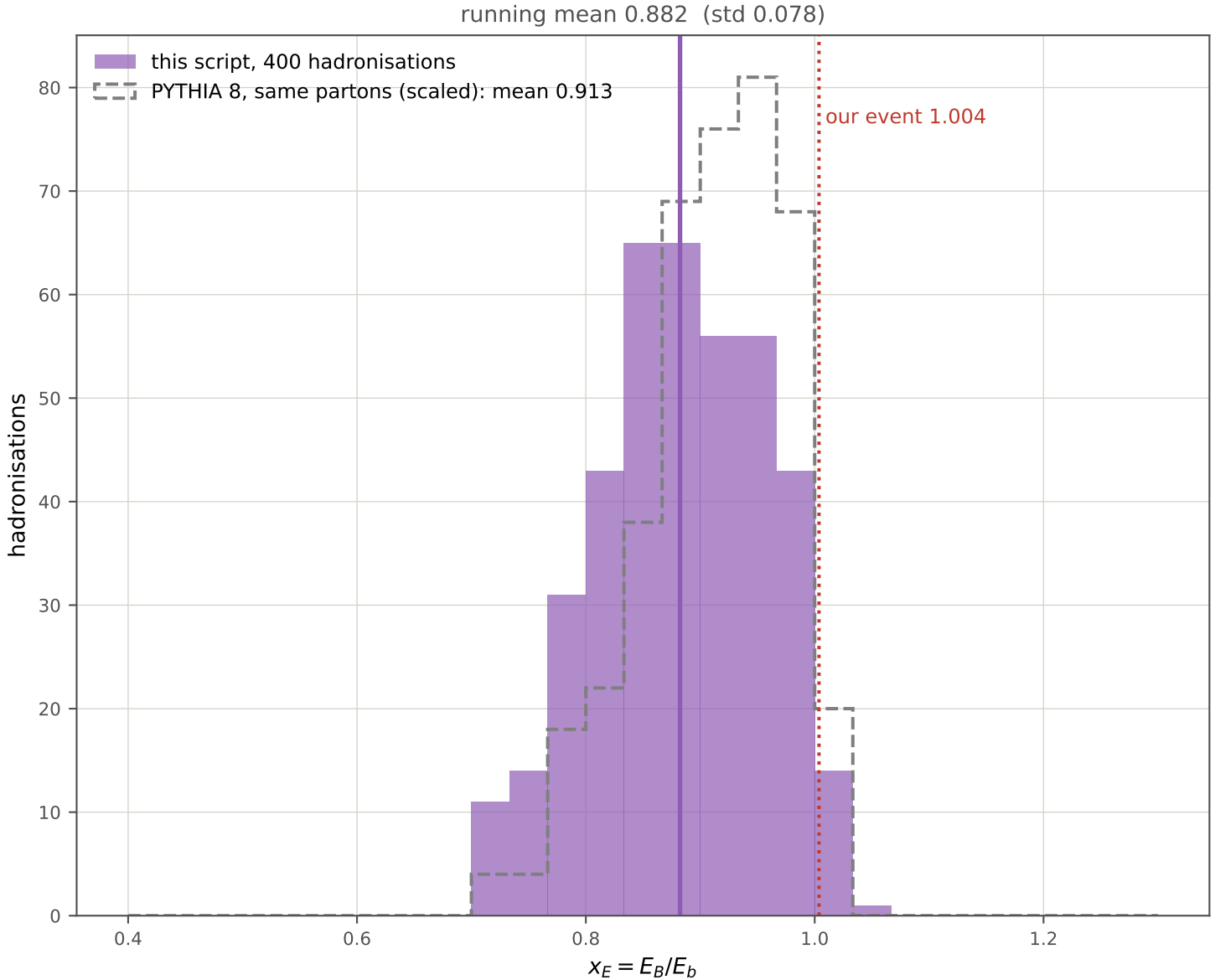
standalone_hadronisation.py

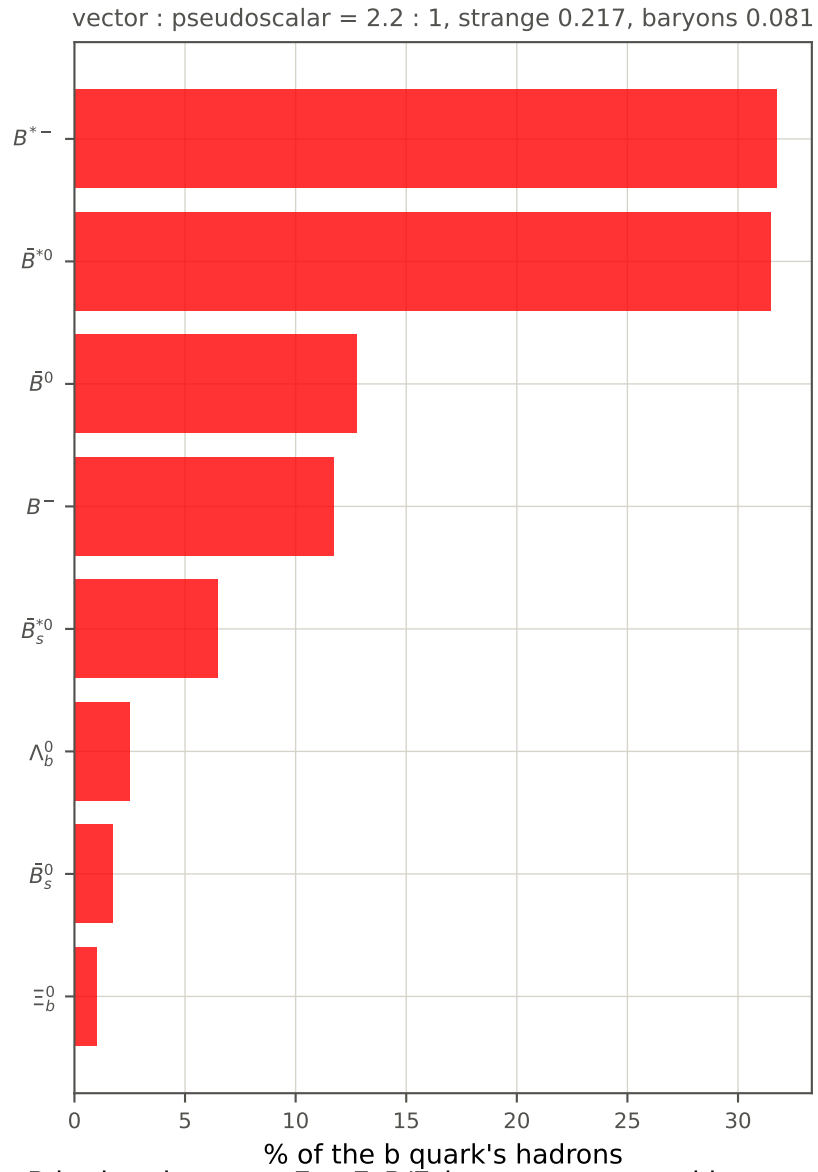


The 211 partons of stage 3 are grouped into 33 colour singlets by their colour tags (4 of them ministrings). String by string, heaviest first, the partons go and the hadrons come: 365 primary hadrons, 184 of them charged, the momentum sum unchanged. The B hadrons of the two b quarks are the red and blue dots.

Scene 9: the same partons hadronised again and again: x_E of the b quark's hadron

standalone_hadronisation.py





The partons of stage 3 are kept and only the hadronisation is redrawn, seeds 1, 2, 3, ...: the energy fraction the B hadron keeps, x_E = E_B/E_b, scatters around its Lund-Bowler mean; PYTHIA 8 on the very same partons (400 times) gives the dashed line. Our event (seed 1) is the first entry. Right: which B hadron it was.